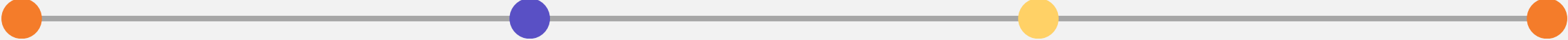
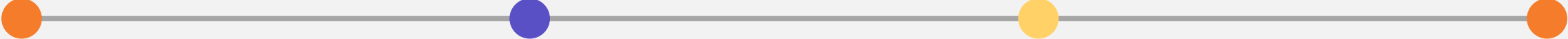




SocBiz Open Project 2026 (Analytics)



**Agentic AI-Based
Dynamic Tariff Optimization
for EV Charging Networks**



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DATA LANDSCAPE & PREPROCESSING

ACN Dataset —Caltech, USA

- 14,848 individual charging sessions
- Apr – Dec 2018 · 54 stations· 1 site
- Avg 9.0 kWh/session and Avg 5.92 hr duration
- Avg idle time: 2.69 hr (charger blocking insight)
- Used for: revenue baseline, pricing efficiency

UrbanEV / ST-EVCDP —Shenzhen, China

- 8,640 timestamps × 247 districts (5-min intervals)
- 30 days ·Jun–Jul 2022
- Occupancy, price, duration, volume per grid
- 57 of 247 districts already on dynamic pricing
- Used for: demand prediction, utilization metrics

KEY PREPROCESSING DECISIONS:

ACN Cleaning

- Dropped 9 empty JSON wrapper columns
- Fixed timestamp format (comma after day)
- Removed sessions > 24 hr & > 100 kWh
- Computed idle time = session – charging duration

UrbanEV Alignment

- Melted wide (8640×247) → long format
- Merged 6 files on timestamp × gridID
- Price multiplier × ₹15 = actual tariff
- Rolling 3-hr volume per grid computed

Feature Engineering

- charger_util_rate = occupancy ÷ count
- queue_length_proxy = max(0, volume/count – 1)
- Pricing zones: surge >80%, discount <30%
- is_peak_hr, CBD, dynamic_pricing flags

KEY EDA FINDINGS & DEMAND BEHAVIOUR INSIGHTS



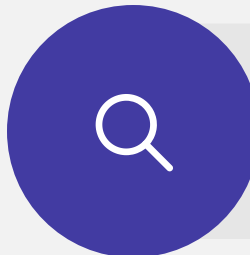
61% Underutilization

61% of UrbanEV readings show util < 30% — discount pricing is the primary lever, not surge



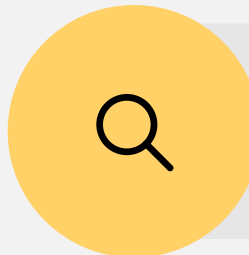
1% Congestion

Only 1% of readings exceed 80% utilization — congestion is rare but high-impact when it occurs



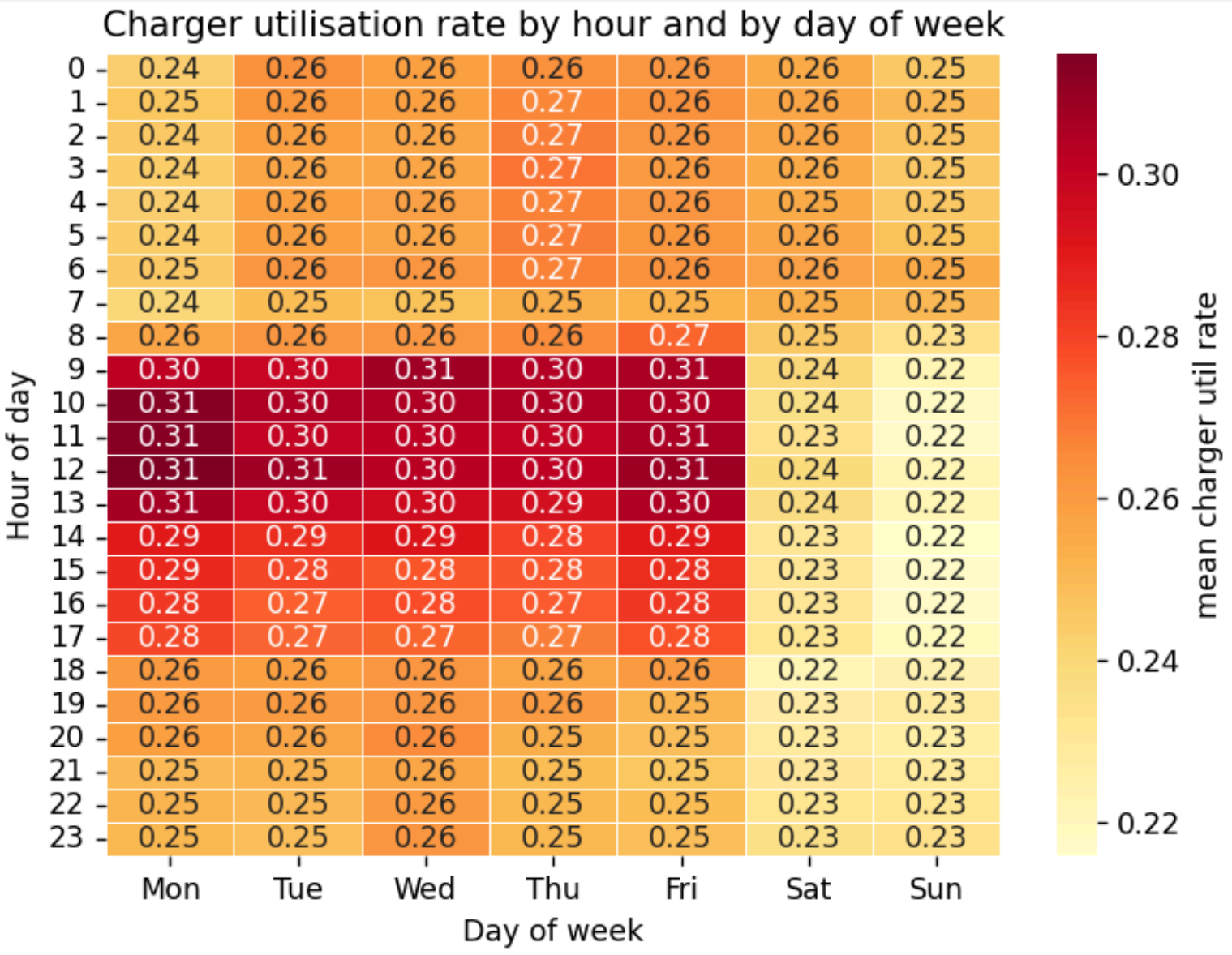
₹9.0 Avg kWh/Session

ACN mean: 9.0 kWh per session. Used as constant in all agent revenue calculations

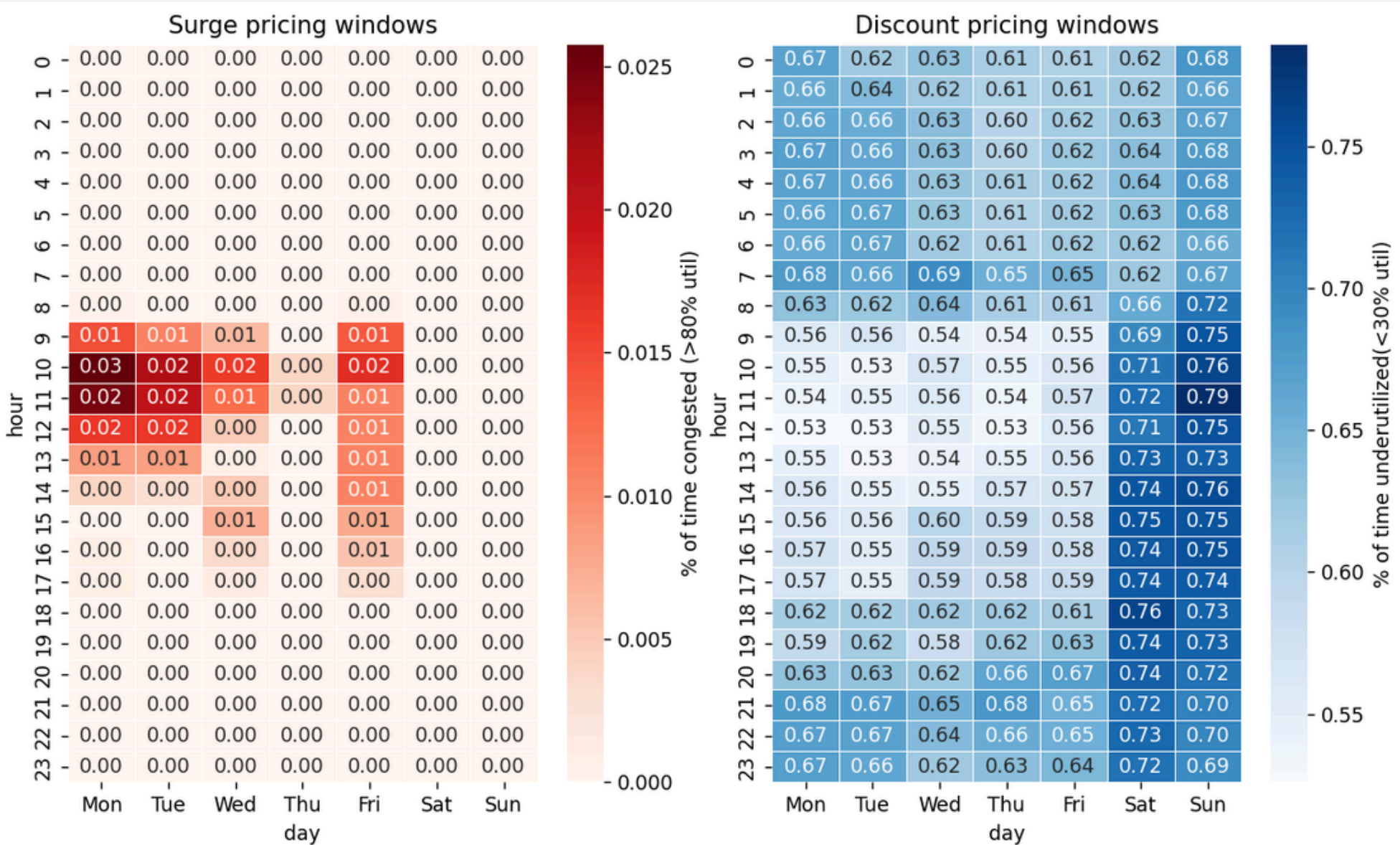


2.69h Avg Idle Time

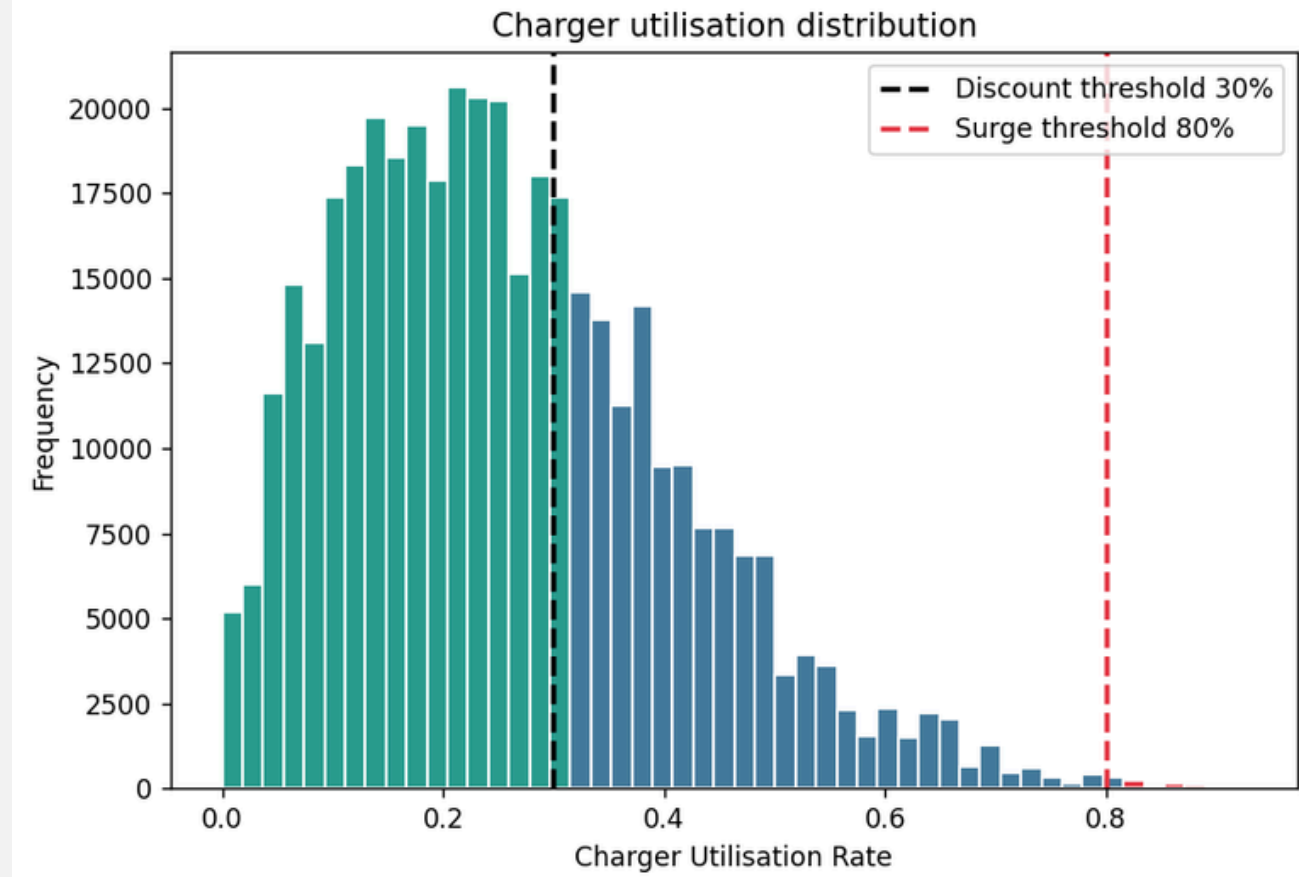
Cars block chargers 2.69 hr after fully charged , idle pricing can free up capacity



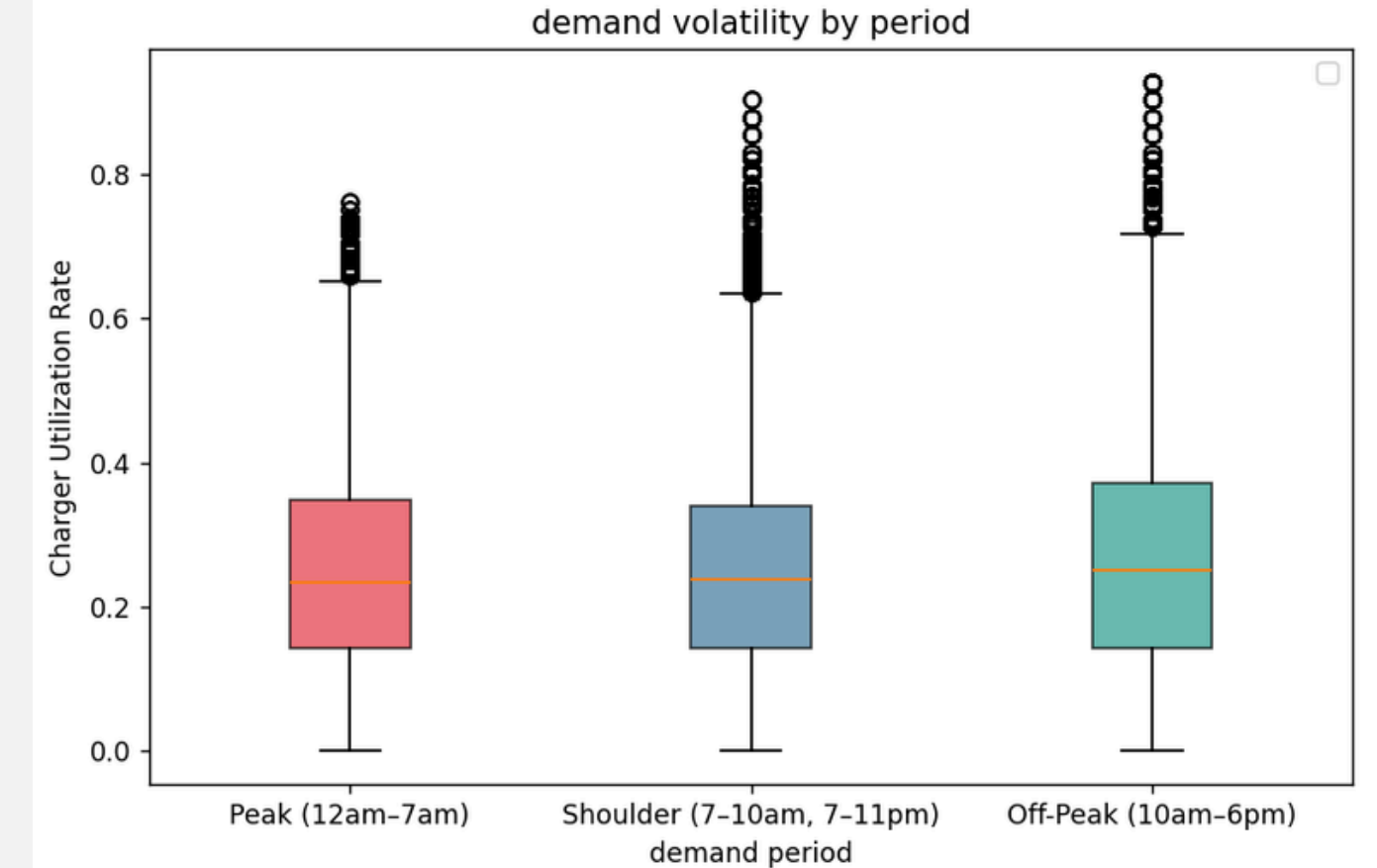
Utilization peaks overnight (midnight–5am) and drops sharply during daytime hours (9am–4pm) directly justifying time-based dynamic pricing windows.



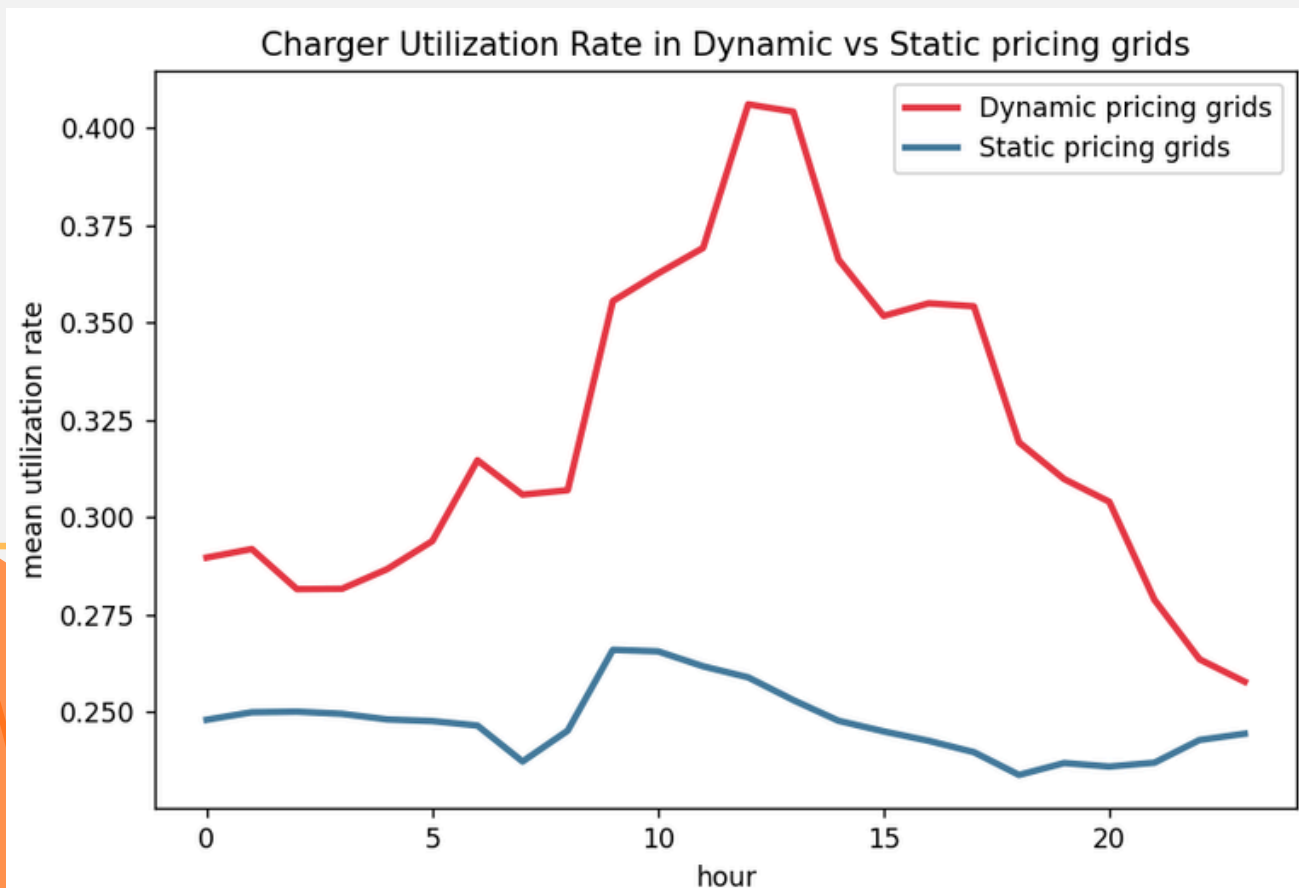
Surge conditions are concentrated in specific hour-day combinations. Discount windows dominate daytime weekdays , making discount pricing the primary revenue lever.



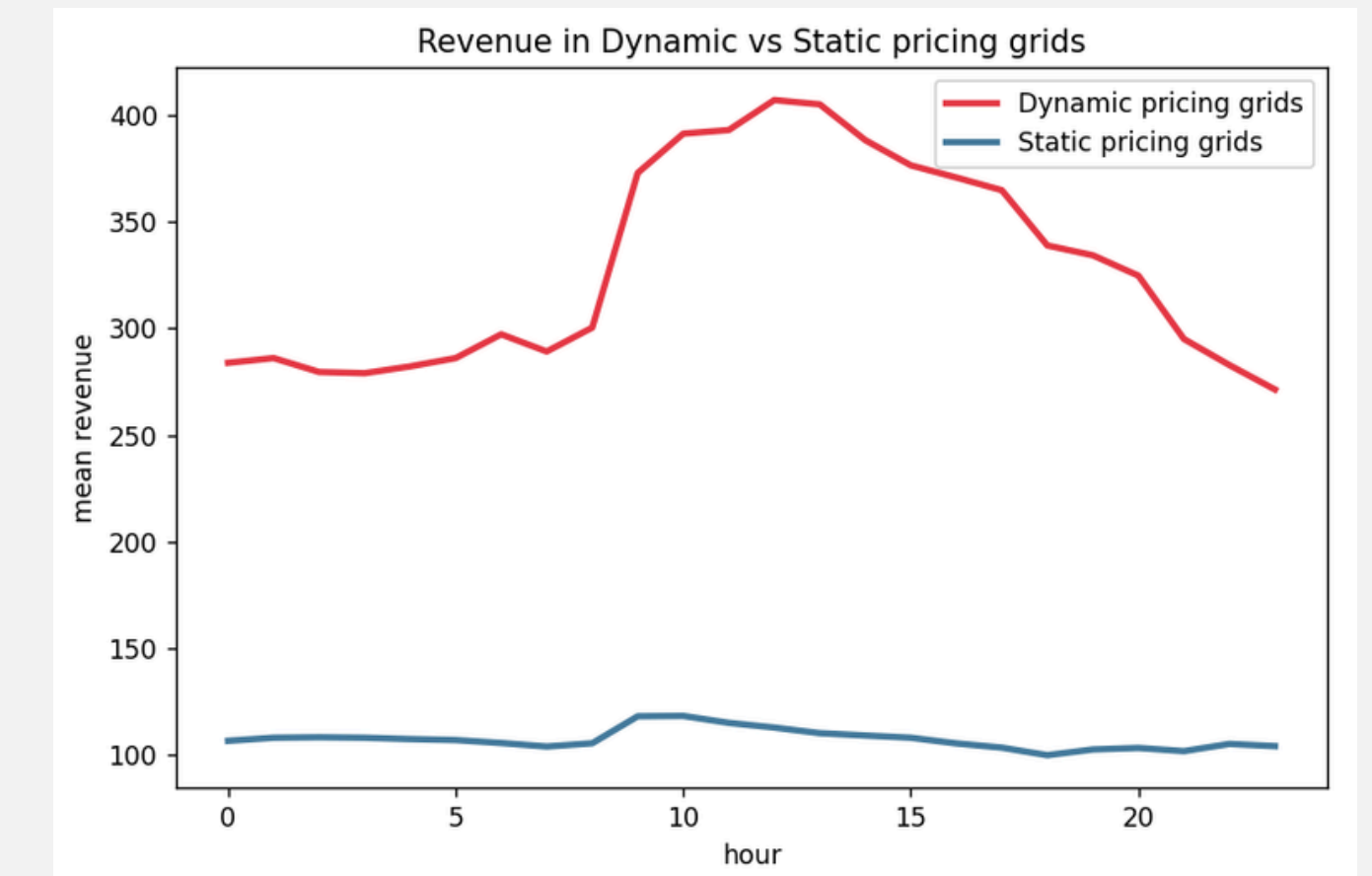
Distribution is heavily right-skewed toward low utilization — confirms the 61% underutilization finding and validates the need for discount-heavy pricing strategy over surge-heavy.



Peak period (midnight-7am) shows the widest spread , high variability means demand prediction is harder and pricing decisions matter more. Off-peak is tightest , stable low demand, consistent discount pricing appropriate.



Districts already on dynamic pricing show measurably different utilization profiles from static ones , direct evidence from within the dataset that dynamic pricing changes demand behaviour



Dynamic pricing associates with higher revenue during peak hours through surge, and comparable revenue off-peak through volume uplift from discounts. Net improvement visible in morning and evening windows.

DEMAND PREDICTION AGENT : MODELLING & RESULTS

>0.85 R^2
Strong fit

~0.04 RMSE
Low error vs 0-1 scale

~0.03 MAE
Mean abs error

XGBoost Regressor

Target Variable:

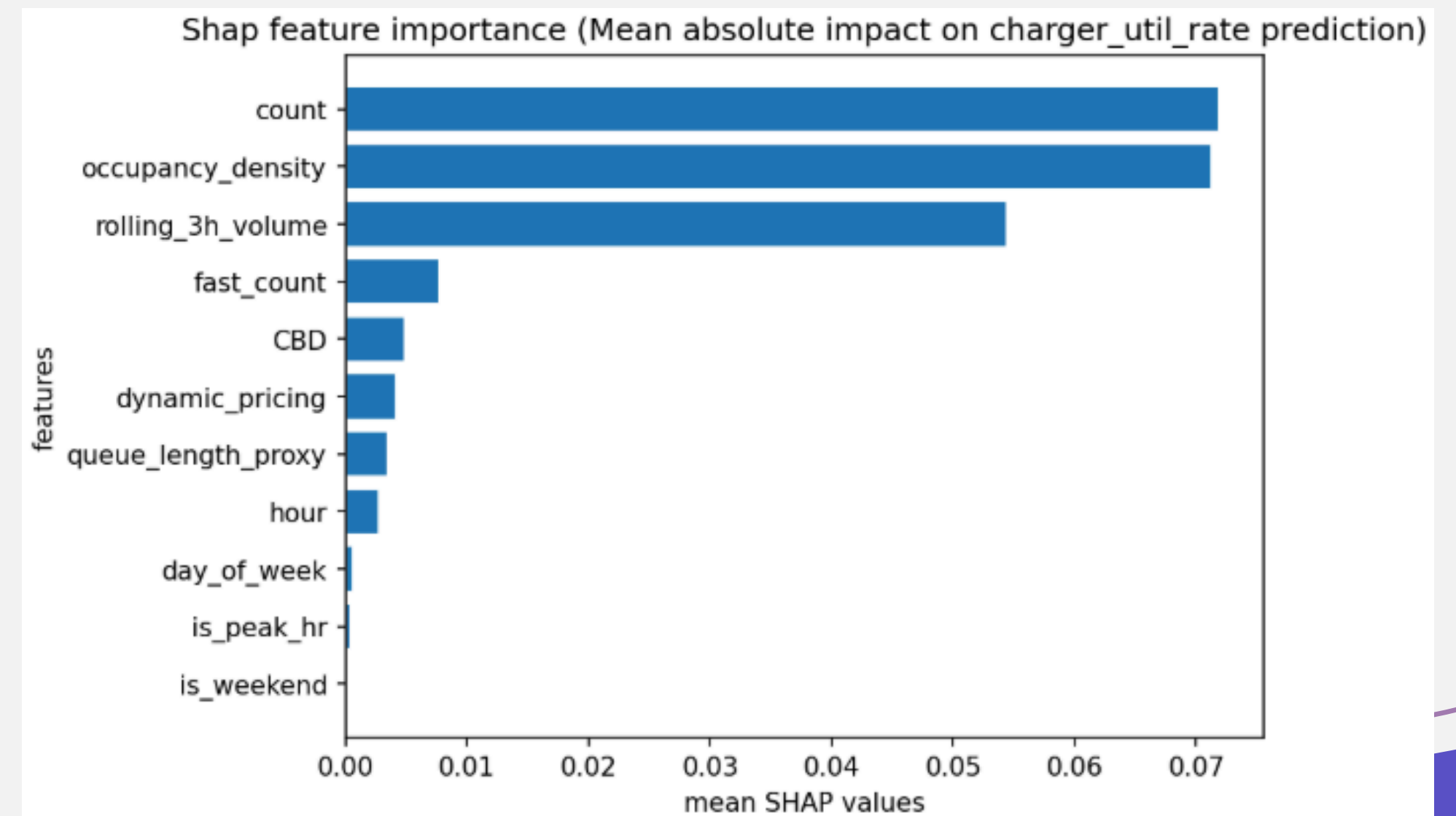
charger_util_rate (occupancy ÷ total chargers)

Features (11)

- hour, day_of_week, is_weekend
- is_peak_hr (18-21h)
- CBD, dynamic_pricing flags
- count, fast_count
- rolling_3h_volume
- occupancy_density, queue_length_proxy

Train-once strategy: model trained on full UrbanEV dataset once. Each episode receives a different time-chunk for prediction — no retrain per episode, ensuring consistent baseline.

SHAP FEATURE IMPORTANCE:



TARIFF PRICING AGENT : OPTIMIZATION LOGIC & OUTCOMES

ACN REVENUE PREDICTION MODEL

- XGBoost trained on ACN sessions
- Target: revenue_per_session (kWh × ₹15)
- Features: hour, session_duration, is_peak, charger_util_rate, queue_proxy, idle_time
- Output: revenue_by_hour lookup (0–23)
- Informs rule-based surge/discount thresholds

$\epsilon = 0.15$ means 15% explore random tariff | 85% exploit highest Q-value tariff

EPSILON-GREEDY MULTI-ARMED BANDIT:

very high > 0.80

Valid tariff range: ₹15–₹24

high 0.60–0.80

Valid tariff range: ₹15–₹24

medium 0.40–0.60

Valid tariff range: Any tariff

low 0.20–0.40

Valid tariff range: ₹8–₹15

very low < 0.20

Valid tariff range: ₹8–₹15

MONITORING AGENT : EVALUATION & FEEDBACK LOOP PERFORMANCE

REWARD SIGNAL DESIGN

- Congested (util>0.80): 40% rev + 60% util reduction
- Underutilized (util<0.30): 20% rev + 80% uplift
- Standard: 100% revenue score

Revenue Gain %

Source: ACN baseline

$(\text{Dynamic Rev} - \text{Static Rev}) / \text{Static Rev} \times 100$

Charger Utilization

Source: UrbanEV

Elasticity-simulated: $\text{util_after} = \text{util_before} \times (1 + \varepsilon \times \Delta P\%/100)$

Off-Peak Uplift

Source: UrbanEV

% increase in sessions in discount-zone grids after pricing

Wait Time Reduction

Source: UrbanEV

Queue proxy reduction $\times$ 5 min per excess session

Customer Response Rate

Source: ACN + $\varepsilon = -0.3$

Price elasticity $\times$ avg price change % across all grids

Pricing Efficiency

Source: ACN kWh

Total dynamic revenue $\div$ total kWh delivered (ACN)

NOTE: Elasticity assumption ($\varepsilon = -0.3$) is a modelling choice, not a measured value. Actual demand response may differ.

BUSINESS, OPERATIONAL & POLICY IMPLICATIONS

Dynamic pricing converts 61% idle/underutilized infrastructure into an active revenue opportunity without adding a single new charger.

Business Value

- Revenue gain vs static ₹15 baseline through surge pricing in congested hours
- Pricing efficiency (₹/kWh) improves as bandit learns better tariff-util mapping
- Off-peak discount pricing attracts sessions to idle infrastructure
- Idle time pricing (future): charge post-charging occupancy to free chargers faster

Operational Impact

- Wait time reduction in surge zones via demand-side management
- Agent memory persists across runs , Q-values improve with each deployment
- Rule-based fallback provides stable pricing when bandit is still exploring
- Time-chunk simulation demonstrates realistic multi-period deployment

Policy & Equity

- Dynamic pricing may disadvantage price-sensitive / lower-income EV users
- Recommended: price caps and discount-hour guarantees for equity protection
- Transparent tariff communication needed before implementation
- Aligns with India's EV infrastructure push under FAME II policy

LIMITATIONS & ASSUMPTIONS:

Elasticity $\epsilon = -0.3$ assumed (not measured) · No live A/B test — elasticity simulation used · ACN and UrbanEV from different countries · ₹15/kWh base tariff adopted per PS, not from data

APPENDIX — ADDITIONAL ANALYSIS & ROBUSTNESS CHECKS

Agent Architecture Detail

- BaseAgent: JSON long-term memory, CSV decision log
- Observe → Think → Act cycle per agent
- Episode count persists across runs via `save_memory()`
- Q-values saved once per episode (not per row)

Preprocessing Robustness

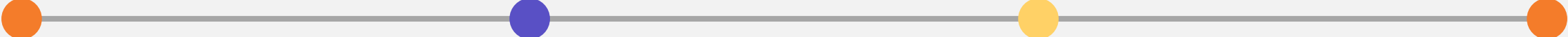
- ACN: 16,304 → 14,848 rows after cleaning
- Outlier removal: >24 hr sessions, >100 kWh
- UrbanEV: 2.13M rows after melt + merge
- Null handling: price nulls filled with multiplier 1.0

Elasticity Sensitivity

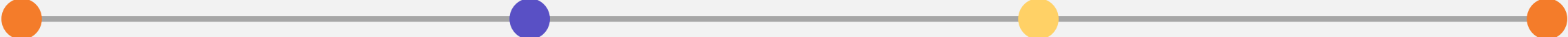
- $\epsilon = -0.3$ chosen as conservative mid-estimate
- Literature range: -0.2 to -0.4 for EV charging
- At $\epsilon = -0.5$: uplift metrics would be ~67% higher
- At $\epsilon = -0.1$: minimal demand shift simulated

Model Validation

- 80/20 train-test split on full UrbanEV data
- SHAP used to validate feature direction (not just rank)
- Revenue model (ACN): RMSE ₹73, R^2 0.53 — moderate fit
- Bandit convergence verified via Q-value printout



THANK YOU!



GITHUB REPO LINK:

https://github.com/meuu09/SocBiz_Open_Project

